README

marrow

The embeddable agent runtime — for robots, edge devices, and native apps where you can't ship Python.

*Reference implementation of **ARI**, the Agent Runtime Interface.*

`marrow` is a native **C++17 agent runtime** — the loop that holds conversation state, calls a model, dispatches tools, and routes messages between agents — with an ergonomic Python binding on top. It is the **reference implementation of ARI**, a language-neutral contract for the agent *runtime* layer.

Why this exists. The mature Python frameworks (LangGraph, CrewAI, AutoGen) are excellent for server apps, but they're Python-locked by design — which rules them out of a fast-growing class of deployments:

- **Embodied / robotic** systems with hard real-time constraints (no GC pauses).
- **Edge / on-device** deployments — constrained hardware, often offline, local models.
- **Native applications** — game engines, audio/DSP, trading systems — that need an agent loop without shipping a Python runtime to the customer.

marrow targets exactly that lane: a native core you can **embed without a managed runtime**, with bounded memory and no hot-path allocation (the **ARI-Embedded** profile). The C++ core compiles as a standalone library; the Python binding is a convenience, not a requirement. Real providers (OpenAI, Anthropic, Ollama) ship as thin Python subclasses that hook back into the core through a trampoline.

ARI sits *beneath* the agent protocol stack — it complements MCP and A2A, it does not compete with them:

```
A2A   — agent-to-agent messaging (across hosts)
MCP   — tool / context exchange (model ↔ world)
ARI   — the runtime that runs a turn   ◀── marrow implements this
        (state · provider · tools · routing)
Model provider (OpenAI / Anthropic / local)
```

See **ARI-SPEC.md** for the contract and **docs/ari-strategy.md** for how the standard is meant to grow.

> **Project status: 0.1.0, public API frozen per `STABILITY.md`.** The C++ core is buildable, tested, and works on Linux / macOS / Windows × Python 3.9–3.12. Production primitives — timeouts, cancellation, bounded inboxes, persistence, tracing, usage tracking — all shipped. Real-world soak testing and bus-factor-of-2 are the remaining items before this should be your default choice for a live system. See `ROADMAP.md`.

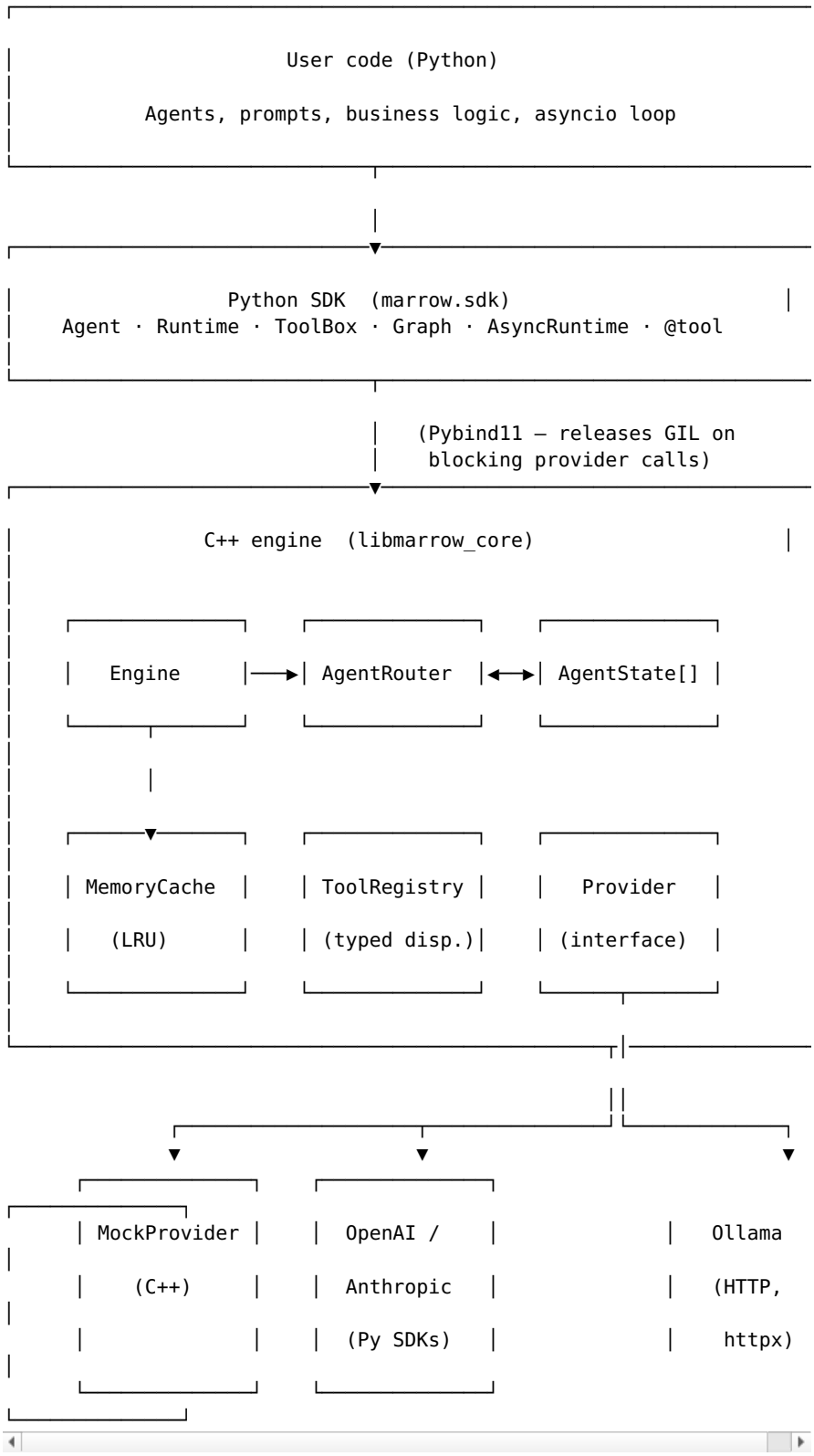
> **Benchmarks** — see `benchmarks/`. M-series Mac, Python 3.12: ~1.4M `AgentState.append`/s, ~672K router send+drain/s, ~64K full 3-agent pipeline iterations/s with `MockProvider`. Run `python -m benchmarks.compare_langgraph` for a head-to-head against LangGraph on a 1-node echo graph (`pip install '.[bench]'`).

Highlights

- **Native C++ core** — Message history, LRU cache, agent router, tool registry, `CancelToken`. C++17, `std::shared_mutex` for read-heavy access, no third-party C++ deps beyond Pybind11.
- **Ergonomic Python** — `Agent` + `Runtime` dataclass API. Two lines from import to your first generated message.
- **Real providers** — `OpenAIProvider`, `AnthropicProvider` (with prompt caching), `OllamaProvider`. Adding a new one is a ~30-line subclass.
- **Streaming first-class** — Every provider implements `generate_stream`. `Agent.stream(on_chunk=...)` works the same way regardless of vendor.
- **Per-call timeouts + cancellation** — Pass `timeout_ms=` and a `CancelToken` to any `step()` or `stream()` call.
- **Backpressure** — `Router.set_inbox_limit(agent_id, max_size, policy)` with three policies (`Reject` / `DropOldest` / `DropNewest`).
- **Graceful shutdown** — `Runtime.shutdown()` blocks new agent creation; in-flight work finishes.
- **Persistence** — `StateStore` interface with `InMemoryStateStore` and `SQLiteStateStore`. Restart your app, resume your conversations.
- **Observability** — `TraceSink` Protocol with `NullTraceSink` / `PrintTraceSink` / `OpenTelemetryTraceSink`. Structured logging via `marrow.logging_config`.
- **Usage tracking + cost estimation** — `UsageTracker` aggregates tokens per agent + model with configurable pricing tables.
- **Retry + rate limiting** — `RetryPolicy` (exponential backoff with jitter) and `RateLimiter` (token bucket); configure once on the `Runtime` and every agent inherits them.
- **Tools dispatched from C++** — Registry lookup and dispatch in native code; tool bodies stay in Python. Tool errors are redacted; args and results have configurable size caps.
- **Fluent graphs, not YAML** — `Graph().start().then().finish()`. Returns a structured `GraphResult` so callers can distinguish "reached end" from "ran out of steps" — no silent truncation.
- **Asyncio bridge** — `AsyncRuntime` wraps the C++ engine in a `ThreadPoolExecutor`. Pybind11 bindings release the GIL on `Provider::generate` and `generate_stream`, so threaded provider calls scale.
- **Thread-safe core** — `Engine`, `AgentRouter`, `AgentState`, `MemoryCache`, `ToolRegistry` verified by `tests/test_concurrency.py`. CI runs the suite under ThreadSanitizer (informational).
- **Embeddable from C++** — The core compiles as a static library independent of Pybind11. See `examples/embed_cpp/` for a pure-C++ demo.
- **Stability promise** — Public API stable per `STABILITY.md` with documented semver + deprecation flow.
- **Tiny dependency footprint** — Core requires only Pybind11. Real

providers are opt-in extras.

Architecture



What lives where

Layer	Responsibilities	Files
C++	Message, AgentState, MemoryCache, AgentRouter,	

core	`ToolRegistry`, `Engine`, `Provider` interface	`src/core/engine.{hpp,cpp}`
Bindings	Pybind11 module, `PyProvider` trampoline, GIL release on hot paths	`src/bindings/bindings.cpp`
Python SDK	`Agent`, `Runtime`, `Graph`, `ToolBox`, `AsyncRuntime`	`python/marrow/`
Providers	`OpenAIProvider`, `AnthropicProvider`, `OllamaProvider` (optional extras)	`python/marrow/providers/`

See `docs/design-notes.md` for the design rationale and deliberate trade-offs.

How marrow compares

	marrow	LangGraph	CrewAI	AutoGen
Native hot path	**C++17**	Python	Python	Python
GIL released during providers	**yes**	partial	no	no
Streaming	**built-in**	yes	yes	yes
Tool dispatch	**C++ registry**	Python	Python	Python
Graph builder	fluent Python	fluent Python	declarative	declarative
Required dependencies	**Pybind11 only**	many	many	many
Multi-agent concurrency	**threads + GIL release**	asyncio	sequential	asyncio
Embeddable from C++	**yes**	no	no	no
Status	pre-alpha	stable	stable	stable

Read this honestly: LangGraph / CrewAI / AutoGen are mature, supported, and have far more features today, and for a server-side Python app you should probably use one of them. `marrow` is not trying to win that comparison on features or on raw speed — if your bottleneck is network I/O (it usually is, server-side), the speedup is modest.

The row that matters is the last one: **Embeddable without a Python runtime.** That is a structural property the others can't offer without ceasing to be Python frameworks — and it's the whole point. If you're putting an agent loop inside a **robot, a drone, an edge device, or a native C++ application**, the Python frameworks aren't an option and marrow is. That lane — the **ARI-Embedded** profile — is what marrow is built to own.

Conformance

marrow is the reference implementation of ARI, and it's held to the spec by an executable **conformance kit** in `ari-conformance/`. Each test maps to a numbered ARI requirement, so "conformant" is

falsifiable, not a marketing claim.

```
        pip install -e ".[test]"
        pytest ari-conformance/ -v          # full kit (needs the built
extension)
        pytest ari-conformance/test_ari_embedded_structural.py -v   #
structural checks, no build
```

The kit runs in CI on every push (`conformance` job). When making an ARI conformance claim, name the profile + version per ARI §10.4, e.g. "*ARI 0.1 · Core + Embedded · kit v0.1*."

Install

```
        python -m venv .venv && source .venv/bin/activate
        pip install -U pip scikit-build-core pybind11 cmake
        pip install -e .
```

With real providers:

```
        pip install 'marrow-rt[openai]'      # OpenAI Chat Completions
        pip install 'marrow-rt[anthropic]'   # Anthropic Messages + prompt
caching
        pip install 'marrow-rt[ollama]'      # local Ollama daemon
        pip install 'marrow-rt[all]'         # everything above
```

The `wheels.yml` workflow builds Linux / macOS / Windows × Python 3.9–3.12 wheels and uploads them as GitHub artifacts on `v*` tag pushes. PyPI publishing is **not** yet wired up; install via git or the artifact downloads until v0.1.

Quickstart

```
        from marrow import Agent, Runtime, MockProvider

        rt = Runtime()
        provider = MockProvider()

        researcher = rt.add(Agent("researcher", provider,
system_prompt="Research."))
        writer     = rt.add(Agent("writer",     provider,
system_prompt="Write."))

        researcher.append_user("Find three facts about graph databases.")
        findings = researcher.step()

        rt.handoff(frm="researcher", to="writer", text=findings)
        rt.deliver(writer)
        print(writer.step())
```

Usage

Real providers

```
        from marrow import Agent, Runtime
        from marrow.providers import AnthropicProvider

        rt = Runtime()
        provider = AnthropicProvider(model="claude-sonnet-4-6")  # caches
system prompt
        agent = rt.add(Agent("assistant", provider, system_prompt="Be
```

```
concise."))

        agent.append_user("What is a graph database?")
        print(agent.step())
```

Streaming

```
        agent.append_user("Write a haiku about graph databases.")
        agent.stream(on_chunk=lambda c: print(c, end="", flush=True))
```

Works identically for `OpenAIProvider`, `AnthropicProvider`, `OllamaProvider`, `MockProvider`, and any Python subclass of `PyProviderBase`.

Tools

```
        from marrow import Runtime, ToolBox, tool

        @tool(description="Add two integers")
        def add(a: int, b: int) -> int:
            return a + b

        rt = Runtime()
        ToolBox().add(add).bind(rt)            # registered in the C++
registry

        print(rt.tools.invoke("add", '{"a": 2, "b": 3}'))
        # {"ok": true, "result": 5}
```

JSON-schema is auto-generated from annotations for the common cases — primitives, `Optional[X]`, `Union`, `list[X]`, `dict`, `Literal[...]`. Stored as a string alongside the tool in C++. For complex types (Pydantic models, dataclasses) pass `schema=` explicitly on `@tool` rather than relying on the auto-generator.

Multi-agent graphs

```
        from marrow import Graph, run_graph

        g = (Graph()
             .start("researcher")
             .then("researcher", "writer")
             .then("writer", "editor", when=lambda s: "DRAFT" in s)
             .finish("editor"))

        final = run_graph(rt, agents_by_name, g, initial_input="Topic:
graphs.")
```

Fluent, introspectable, no YAML. See design-notes for why.

Async / concurrency

```
        import asyncio
        from marrow import AsyncRuntime, Agent, MockProvider, Runtime

        async def main():
            rt = AsyncRuntime(Runtime())
            agents = [rt.add(Agent(f"w{i}", MockProvider())) for i in
range(8)]
            for i, a in enumerate(agents):
                a.append_user(f"hello from worker {i}")
            results = await rt.gather_steps(agents)   # concurrent

        asyncio.run(main())
```

The C++ binding releases the GIL during `Provider::generate`, so threaded steps make real parallel progress while their Python provider code runs.

Examples

All runnable. From the repo root after `pip install -e .`:

File	Demonstrates
examples/main.py	Two-agent handoff
examples/tools_example.py	Tool registry, schema generation, invocation
examples/graph_example.py	Three-node graph execution
examples/streaming_example.py	Word-by-word streaming
examples/async_example.py	Concurrent agent steps
examples/openai_example.py	OpenAI provider (requires `OPENAI_API_KEY`)
examples/anthropic_example.py	Anthropic provider with prompt caching
examples/embed_cpp/	Embedding the C++ core into a non-Python application

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Tests

```
pip install -e ".[test]"
pytest -v
```

Current suite: ~20 tests across smoke, router edge cases, tool registry (including args/result size caps and error redaction), streaming (Python-subclassed `Provider` trampoline path), and a dedicated concurrency suite that exercises the `shared_mutex` paths under contention.

Project status

This is **0.1.0** — the public API is frozen per STABILITY.md. The C++ core builds and runs on Linux / macOS / Windows × Python 3.9–3.12. The architecture has been pressure-tested and the controversial trade-offs are documented in docs/design-notes.md.

Against the ARI profiles, marrow currently targets **ARI-Core** and **ARI-Embedded** (with **ARI-Server** features present but not yet soak-tested).

What is shipped and stable:

- Core message / state / cache / router APIs (`std::shared_mutex`, read-heavy)
- Tool registration via `@tool` + `ToolBox`, dispatched from the C++ registry
- Graph builder with structured `GraphResult`
- `AsyncRuntime` for concurrent steps (GIL released during provider calls)
- Mock + OpenAI + Anthropic + Ollama providers (best-effort, not battle-tested)
- Per-call timeouts + `CancelToken` on `step()` / `stream()`
- Bounded router inboxes with `Reject` / `DropOldest` / `DropNewest` policies

- Persistence — `StateStore` with in-memory + SQLite backends
- Observability — `TraceSink` with null / print / OpenTelemetry sinks
- Usage tracking, retry, and rate-limiting configured on the `Runtime`
- Embeddable pure-C++ core (see `examples/embed_cpp/`)

What is **not** done and should not yet be relied on:

- Real-world soak testing and a bus factor ≥ 2 (the remaining gates before this is your default for a live system)
- Full multi-turn tool-use protocol (automatic tool-call loop inside `step`)
- Published, *measured* benchmarks vs LangGraph (the harness exists; numbers are not yet committed)
- PyPI publishing — wheels build to GitHub artifacts; trusted publishing is wired but nothing is on PyPI yet
- ABI stability across versions (source-level stability only, per `STABILITY.md`)
- An **ARI conformance kit** — the executable proof of conformance (see `docs/ari-strategy.md`)

Roadmap

- **v0.1** — Real provider tests w/ key-gated CI; ToolCall integration in the step loop; streaming benchmark vs LangGraph
- **v0.2** — `StateStore` interface + SQLite impl; `TraceSink` for OTel; bounded inboxes
- **v0.3** — DAG executor; async-iterator streaming API; first stable API contract
- **v1.0** — ABI promise per module; cibuildwheel wheels on PyPI; plugin loader for C++ providers

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`marrow` is licensed under the **Apache License, Version 2.0**. The full text is in `LICENSE` and required attribution is in `NOTICE`.

Why Apache 2.0?

Picked over MIT specifically for the **explicit patent grant in §3** — contributors grant a perpetual, worldwide, royalty-free patent license to users for any patents reading on their contributions. This protects downstream commercial adopters from patent ambushes by contributors. Picked over GPL because `marrow` is intended to be embedded in proprietary products without forcing them to be open-sourced.

What the license permits

- **Commercial use** — including embedding in proprietary or closed-source products, no royalty
- **Modification** and creation of derivative works
- **Distribution** of source and binary forms
- **Private use** without disclosure
- **Sublicensing** under different terms (e.g. a downstream MIT package can include marrow)

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- **Carry forward** the contents of `NOTICE` in derivative distributions, where applicable

- **State modifications** if you redistribute a modified version (Apache §4(b))
- **Not use** "marrow" as the primary brand of a competing or derivative product (see *Trademark* below). The Apache license does *not* grant trademark rights.

What you do not get

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- Liability — contributors are not liable for damages from use (Apache §8)
- Trademark license — see below

Trademark

The name "marrow" is not currently a registered trademark. We ask the community to:

- **Welcome:** use the name to describe integrations, plugins, or compatible implementations ("X for marrow", "marrow-compatible Y")
- **Avoid:** using "marrow" as the primary brand of a competing or derivative product to prevent user confusion

Patent statement

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Contributing

Pull requests are welcome. See `CONTRIBUTING.md` for development setup, code style, and how submissions are licensed.

By submitting a PR you confirm — under Apache §5 — that your contribution is licensed to the project under the same Apache 2.0 terms, without additional restrictions.

Acknowledgements

- Pybind11 — the seamless C++/Python interop that makes this whole project tractable.
- scikit-build-core — modern CMake-driven Python builds.
- cibuildwheel — wheels for every platform with one workflow.

Inspiration from LangGraph, CrewAI, AutoGen, and the broader open-source agent ecosystem — `marrow` is an alternative, not a replacement.
